

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit 3 : Session 11 : SLO 2 — IEEE 754 Double Precision Format

■ Section A: Match the Following (Remember)

- A. Total bits in double precision → 3. 64 bits
- B. Sign bit → 4. Indicates positive or negative
- C. Exponent field → 1. 11 bits
- D. Mantissa (fraction) → 2. 52 bits
- E. Exponent bias in double precision → 5. 1023

■ Section B: Fill in the Blanks (Understand)

- Double precision format uses a total of 64 bits.
- The exponent field in double precision is 11 bits long.
- The exponent is stored with a bias of 1023 in double precision format.
- The mantissa is composed of 52 bits and represents the fractional part.
- The sign bit is set to 0 for positive numbers and 1 for negative numbers.

■ Section C: Activity — Encode -25.625 in IEEE 754 Double Precision

- Convert 25.625 to binary: $25 = 11001$, $0.625 = 0.101 \rightarrow 11001.101$
- Normalize: 1.1001101×2^4
- Sign bit: 1 (negative number)
- Biased exponent: $4 + 1023 = 1027 = 10000000011$
- Mantissa (52 bits): 1001101 000 (padded with zeros)
- Full 64-bit representation: $1\ 10000000011\ 100110100$